

Gyanendra Chaubey

AI/ML Engineer | M.Tech IIT Jodhpur | Agentic AI & Deep Learning

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PROFILE SUMMARY

AI/ML Engineer with 3+ years of industry and applied research experience designing production-grade agentic AI systems, multimodal RAG pipelines, and deep learning models across healthcare, e-commerce, and mental wellness. Published researcher with peer-reviewed work in IEEE Access (Q1) and Springer.

TECHNICAL SKILLS

Languages: Python, SQL, C++

AI/ML Stack: PyTorch, TensorFlow, HuggingFace, scikit-learn, XGBoost, FAISS, Qdrant, ColPali, Pandas, NumPy, Vector Databases, Model Evaluation

GenAI & Agentic AI: LangGraph, LangChain, LlamaIndex, RAG (multimodal), Fine-tuning, RLHF, Diffusion Models, LLMs, Groq, Prompt Engineering, MCP, CrewAI, MLflow

Deployment & Cloud: FastAPI, Docker, Kubernetes, Nginx, React (TypeScript), AWS, GCP, Snowflake, BigQuery, Git, CI/CD

NLP & Computer Vision: BERT, Semantic Search, Transformers, ViT, CNN, Object Detection, Image Segmentation

PROFESSIONAL EXPERIENCE

Brainwave Science

September 2025 – Present

AI Research Engineer

Delhi

- Architected and deployed a production-grade multi-agent mental wellness platform (**Calming AI**) using **LangGraph StateGraph**, orchestrating 3 specialized LLM agents (emotional assessment, wisdom therapy, video recommendation) with a 5-layer intent pipeline (9 intent types, LLM primary + keyword fallback) and hybrid crisis detection — achieving sub-second routing latency and 95% intent-shift detection accuracy across multi-turn conversations; deployed to production serving real users across English, Hindi, Hinglish, and Spanish.
- Engineered a dual **FAISS-HNSW RAG** system indexing 13,000+ Bhagavad Gita verses and 200+ therapy PDFs using **multilingual-e5-large** (1024-dim) embeddings with symptom-aware score boosting, LLM re-ranking (Groq Llama 3.3-70B), and rolling conversation summaries every 5–10 turns to preserve long-context therapeutic state; similarity score above 90 and groundedness up to 80.
- Built a multilingual voice I/O pipeline using Groq Whisper STT with 3-way VAD decision logic (ACCEPT/REJECT/AMBIGUOUS with Silero tiebreaker) and multi-provider TTS (Microsoft Edge TTS + ElevenLabs fallback) with emotion-aware prosody; deployed full stack via **FastAPI + React 18 (TypeScript) + Docker + Nginx**.

iHub and HCI Foundation, IIT Mandi

June 2025 – August 2025

Senior Associate (Multimodal AI & ML Research)

Himachal Pradesh

- Engineered a multimodal RAG pipeline using **ColPali** for cross-modal retrieval across 1,000+ page technical PDFs, integrating **Qdrant** and **Supabase** as vector stores for scalable, low-latency document search.
- Designed and trained a CNN-based deep learning model for automated textile defect classification (neps, slubs, thick-thin variations) on an imbalanced production dataset, achieving ~80% accuracy.
- Built a multimodal data quality and curation pipeline for EEG, audio, and video streams, defining annotation standards and automated validation checks for a research-grade dataset.

HCL Technologies Ltd

April 2021 – July 2023

Software Engineer

Lucknow

- Developed, trained, and deployed supervised ML models (**XGBoost**, time-series) for next-purchase prediction in a production e-commerce system, improving forecasting accuracy by 5%, directly informing inventory planning and personalised recommendations for millions of customers; served predictions via REST API integrated into the customer data platform.
- Designed end-to-end ML data preprocessing pipelines — feature engineering, outlier detection, and class-imbalance treatment — across millions of records, achieving 98%+ data quality and reducing model training error rate.
- Engineered and maintained 50+ **ETL** pipelines using **SSIS** and **Python**, automating data ingestion and transformation into **Snowflake** and **Google BigQuery**, enabling reliable analytics and model retraining workflows.

PROJECTS

RibCageImp: A 3D Deep Learning Framework for Rib Cage Implant Generation (M.Tech Project)

04/2025

Supervised by: Dr. Deepak Mishra

- Constructed a custom training dataset from RibFrac CT scans by artificially introducing rib defects and generating corresponding implant labels. Trained UNet and diffusion model architectures to predict 3D implant geometry directly from CT input. [Project](#) | [GitHub](#)

Predicting Multiple Product Features from Images for Enhanced E-Commerce Cataloging10/2024

Mentored by: Dr. Mayank Vatsa

– Proposed a dual-backbone architecture using ViT-H/14-quickgelu (DFN-5B) and ConvNext-XXLarge (LAION-2B) for predicting product attributes (color, pattern, sleeve length) from images. MLP heads specialized by (category, attribute) pairs enable a score of 0.802 with just 0.05s/image on an NVIDIA T4 GPU. [GitHub](#)

EDUCATION

Indian Institute of Technology, Jodhpur2023–2025

M.Tech. in AI & Data Science (AR-VR) – CGPA: 8.55India

Rajkiya Engineering College, Banda2016–2020

B.Tech. in Information Technology – CGPA: 8.4India

PUBLICATIONS

T2I-BiasBench: A Multi-Metric Framework for Auditing Demographic and Cultural Bias in Text-to-Image Models2026

Jaiswal, N., Arjaria, S., **Chaubey, G.**, Kumar, A., Singh, A., Chaurasiya, A. | *arXiv:2604.12481 (preprint)* | [Paper](#)

Genetic Algorithm-Based Search Space Exploration to Generate Best Convolutional Neural Network2025

Bisen, D., Saurabh, P., Thakur, M., **Chaubey, G.**, Singh, U., Dubey, A. | *IEEE Access (peer-reviewed, Q1 journal)* | [Paper](#)

RibCageImp: A Deep Learning Framework for 3D Ribcage Implant Generation2024

Chaubey, G., Farooq, A., Singh, A., Mishra, D. | *arXiv:2411.09204 (preprint)* | [Paper](#)

Erythemato-Squamous Diseases Prediction and Interpretation Using Explainable AI2024

Rathore, A. S., Arjaria, S., Gupta, M., **Chaubey, G.**, et al. | *IETE Journal of Research, 70(1) (peer-reviewed)* | [Paper](#)

Customer Purchasing Behavior Prediction using Machine Learning Supervised Methods2023

Chaubey, G., Gavhane, P.R., Bisen, D., Arjaria, S. | *Springer JAIHC (peer-reviewed, Q2 journal)* | [Paper](#)

POSITIONS OF RESPONSIBILITY & ACHIEVEMENTS

2nd Prize Winner, AVINYA Hackathon, ICMI Conference, IIT Jodhpur2024

PG Representative, IIT Jodhpur2024-25

Institute Branch Topper, Department of Information Technology, REC Banda2020

CERTIFICATIONS

NPTEL-IIT Madras: ML, DBMS, Data Science for Engineers (78%, Silver medal); Python for Data Science (Top 5%, Silver medal); Data Mining

AWS: Certified Cloud Practitioner